

CMPE 220 System Software: CPU Design

Professor: Ishie Eswar

Team Members (Group 6):

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GitHub Repository

GitHub Repository Link: <https://github.com/ParthPatel00/CPU-Design>

Git Clone Link: <https://github.com/ParthPatel00/CPU-Design.git>

How to Download, Compile, and Run

Requirements:

- macOS or Linux with GCC installed
- Git
- Make

Step 1: Download

```
git clone https://github.com/ParthPatel00/CPU-Design.git
```

```
cd CPU-Design/src
```

Step 2: Build

```
make clean && make
```

Cleans and rebuilds everything. Creates two executables: cpu (the emulator) and asm (the assembler).

Step 3: Assemble and Run the Recursive Multiply Program

```
./asm ../programs/multiply.asm ../programs/multiply.bin
```

```
./cpu run ../programs/multiply.bin
```

Expected output: 6. This runs multiply(3, 2), which computes $3 \times 2 = 6$ using recursion.

Step 4: Watch the Full Execution Trace

```
./cpu verbose ../programs/multiply.bin
```

This prints all 39 cycles showing every FETCH, DECODE, and EXECUTE stage with the complete CPU state after every instruction.

Team Member Contributions

Parth Patel:

Designed overall CPU architecture, instruction set, instruction encoding, and memory map. Implemented all emulator modules: `cpu.c` (register file), `alu.c` (arithmetic and flag logic), `control.c` (fetch/decode/execute loop), `bus.c` (memory routing), `memory.c` (64K word array and binary loader), `io.c` (memory-mapped IO devices), and `main.c` (CLI entry point). Created CPU architecture documentation and schematic.

Harshitha Vadavalli:

Implemented two-pass assembler (`assembler.c`, `asm_main.c`). Wrote all four assembly programs: `timer.asm`, `hello.asm`, `fibonacci.asm`, and `multiply.asm`. Created Makefile build targets. Wrote both project reports. Wrote README and project documentation. Managed GitHub repository.